Deep Reinforcement Learning for Market Making in Hawkes-Driven Markets: an Event-Time Formulation of the Cartea–Jaimungal–Ricci Model

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October 28, 2026

Abstract

We embed the bivariate sum-of-exponentials Hawkes model of market orders of our previous work into the market-making framework of Cartea, Jaimungal and Ricci, and derive the Hamilton–Jacobi–Bellman equation and the optimal quotes. Following Guéant and Manziuk, we rewrite the problem in event time. Unlike in their setting, the state moves between two market orders, so the Bellman equation involves an integral along a deterministic flow, with an explicit survival function for the waiting time. The associated operator is a contraction, and its sampled version is used as the critic target of TD3 and SAC agents. The agents are trained in the calibrated simulator and evaluated on 22 DOTUSDT sessions against Avellaneda–Stoikov. All methods lose money after fees under the primary fill rule, and the differences between methods are not significant.

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1 Introduction

A market maker posts bid and ask limit orders and earns the spread when they are hit by market orders. It bears two risks: inventory risk, since each execution moves its position away from zero, and adverse selection, since the price tends to rise after a buy order, just after the market maker has sold. The models of Ho and Stoll [16] and Avellaneda and Stoikov [1], later solved by Guéant, Lehalle and Fernandez-Tapia [11] and Guéant [10], treat the inventory problem with Poisson arrivals. They do not account for the clustering of market orders observed in our previous paper [28].

Cartea, Jaimungal and Ricci [6] (CJR hereafter) model market orders by a mutually exciting Hawkes process [15, 2], each order also moving a short-term drift of the price (the "alpha"). The optimal quotes then depend on the whole state of the order flow, and with a sum-of-exponentials kernel the HJB equation has too many dimensions for grid methods.

Guéant and Manziuk [12] met the same problem for corporate bonds and used an actor–critic reinforcement learning (RL) algorithm. They rewrite the problem in event time: decisions only matter when a request arrives, so the value function can be learned from transitions between requests. In their model the state does not move between two requests and the discounting has a closed form. In ours, the intensities and the alpha decay between market orders, so the event-time equation requires an integral along the flow.

We derive the HJB equation of the CJR problem with the order flow of [28] and the optimal quotes (Section 3), then its event-time Bellman equation, which is a contraction and reduces to a finite system in the Poisson case (Section 4). We describe how TD3 [9] and SAC [13] learn it as a semi-Markov decision process with a random discount factor (Section 5), and evaluate them on DOTUSDT against Avellaneda–Stoikov (Section 6).

The reader may proceed directly to the empirical results in Section 6.

2 Model and Problem Formulation

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{R}_+}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. Throughout, prices and quote distances are expressed in basis points of a fixed reference mid-price, and quantities in lots (see Section 6 for the units used in the implementation).

2.1 Order flow

Let M_t^+ (resp. M_t^-) denote the number of buy (resp. sell) market orders arrived up to time t. Following [28], the pair (M^+, M^-) is a bivariate Hawkes process (Definition 2.1).

Definition 2.1 (Bivariate order flow)**.** The conditional intensities of (M^+, M^-) are

$$\lambda_t^i = \mu_i + \sum_{j\in\{+,-\}} \int_{[0,t)} \varphi_{ij}(t-s)\,\mathrm{d}M_s^j, \qquad \varphi_{ij}(t) = \sum_{p=1}^{P} a_{ij}^{(p)}\, e^{-\beta_{ij}^{(p)} t}, \qquad i \in \{+,-\},$$

with $\mu_i > 0$, $a_{ij}^{(p)} \geq 0$, $\beta_{ij}^{(p)} > 0$ and $P = 2$. We assume that the branching matrix $\Gamma_{ij} := \sum_p a_{ij}^{(p)}/\beta_{ij}^{(p)}$ satisfies $\rho(\Gamma) < 1$ (stationarity, [28, Prop. 2.14]).

Remark 2.2 (Notation). In [28] the kernel amplitudes were denoted $\alpha_{ij}^{(p)}$. We write them $a_{ij}^{(p)}$ here, because α denotes the price drift in the CJR framework, and we keep η for the branching ratio $\eta = \rho(\Gamma)$.

As in [28], the intensities are Markovian once we introduce the *excitation states*

$$R_{i,j}^{(p)}(t) := \int_{[0,t)} e^{-\beta_{ij}^{(p)}(t-s)}\,\mathrm{d}M_s^j, \qquad \lambda_t^i = \mu_i + \sum_{j\in\{+,-\}} \sum_{p=1}^{P} a_{ij}^{(p)} R_{i,j}^{(p)}(t),$$

where $R^{(p)}_{i,j}$ decays at rate $\beta^{(p)}_{ij}$ between events and jumps by $+1$ at each event of type j ([28, Prop. 4.4]). Its first index is the excited side (target), its second the exciting side (source).

2.2 Price, alpha and fill probabilities

The mid-price S follows

$$\mathrm{d}S_t = \alpha_t \,\mathrm{d}t + \sigma \,\mathrm{d}W_t,$$

where W is a standard Brownian motion and α is a short-term drift driven by the order flow [6]:

$$\mathrm{d}\alpha_t = -\zeta\, \alpha_t \,\mathrm{d}t + \sigma_\alpha \,\mathrm{d}B_t + \epsilon^+ \,\mathrm{d}M^+_t - \epsilon^- \,\mathrm{d}M^-_t,$$

with $\zeta > 0$, $\epsilon^\pm \geq 0$, $\sigma_\alpha \geq 0$ and B a Brownian motion independent of W and of (M^+, M^-). A buy order pushes the drift up, a sell order pushes it down, and the effect decays at rate ζ.

The market maker posts an ask at $S^a_t = S_t + \delta^+_t$ and a bid at $S^b_t = S_t - \delta^-_t$, where $\delta^\pm_t$ are $\mathcal{F}_{t^-}$-measurable (predictable). Buy market orders can only lift our ask and sell market orders can only hit our bid; they play the role of the requests for quotes of [12]. Conditionally on the arrival of a buy (resp. sell) market order, our ask (resp. bid) is filled with probability $h_+(\delta^+_t; \kappa^+_{t^-})$ (resp. $h_-(\delta^-_t; \kappa^-_{t^-})$), independently of everything else given the current state. The parameter $\kappa^\pm$ measures the depth of market orders; in CJR it is itself stochastic,

$$\mathrm{d}\kappa^\pm_t = \beta_\kappa(\theta_\kappa - \kappa^\pm_t)\,\mathrm{d}t + \eta_\kappa \,\mathrm{d}M^\pm_t + \nu_\kappa \,\mathrm{d}M^\mp_t,$$

Assumption 2.3 (Fill probability)**.** For every $\kappa > 0$, $\delta \mapsto h_\pm(\delta;\kappa)$ maps $\mathbb{R}_+$ to $(0,1]$ and satisfies

(H1) $h_\pm(\cdot\,;\kappa)$ is C^2 on $\mathbb{R}_+$;

(H2) $h_\pm(\cdot\,;\kappa)$ is strictly decreasing: $h'_\pm(\delta;\kappa) < 0$;

(H3) $\lim_{\delta\to+\infty} \delta\, h_\pm(\delta;\kappa) = 0$;

(H4) $\displaystyle\sup_{\delta\geq 0} \frac{h_\pm(\delta;\kappa)\, h''_\pm(\delta;\kappa)}{\big(h'_\pm(\delta;\kappa)\big)^2} < 2.$

These are the assumptions of Guéant [10], written for a probability; derivatives are taken in δ.

Example 2.4 (Exponential fill probability)**.** The CJR choice $h_\pm(\delta;\kappa) = e^{-\kappa\delta}$ satisfies Assumption 2.3: (H1)–(H3) are immediate and $hh''/(h')^2 = \kappa^2 e^{-2\kappa\delta}/(\kappa^2 e^{-2\kappa\delta}) = 1 < 2$.

Remark 2.5 (Implemented specification). In our implementation, $h_\pm(\delta;\kappa) = e^{-\kappa\delta}$ with a constant κ ($\eta_\kappa = \nu_\kappa = 0$, $\kappa^\pm_0 = \theta_\kappa = \kappa$), the drift is pure-jump ($\sigma_\alpha = 0$), and the decay rates depend only on the kernel component, $\beta^{(p)}_{ij} = \beta^{(p)}$. In the latter case $R^{(p)}_{+,j} = R^{(p)}_{-,j}$, so the excitation state reduces to the $2P = 4$ variables $R^{(p)}_j$, $j \in \{+,-\}$. The derivations below hold in the general case unless stated otherwise.

2.3 Inventory, cash and P&L

Let N^+_t (resp. N^-_t) count the executions of our ask (resp. bid) on $[0,t]$, and let $\Delta > 0$ be the quote size. The inventory takes values in the grid $\mathcal{Q} = \{-Q, -Q+\Delta, \ldots, Q-\Delta, Q\}$ and follows

$$\mathrm{d}q_t = \Delta\,\mathrm{d}N^-_t - \Delta\,\mathrm{d}N^+_t.$$

The market maker stops quoting on the ask when $q_{t^-} = -Q$ and on the bid when $q_{t^-} = Q$. Since each market order fills our quote with probability $h_\pm$, $N^\pm$ only jump at jump times of $M^\pm$, with intensity

$$\Lambda^+_t = \lambda^+_t\, h_+(\delta^+_t; \kappa^+_{t^-})\, \mathbb{1}_{\{q_{t^-} > -Q\}}, \qquad \Lambda^-_t = \lambda^-_t\, h_-(\delta^-_t; \kappa^-_{t^-})\, \mathbb{1}_{\{q_{t^-} < Q\}}.$$

Each execution pays a maker fee $c \geq 0$ per unit, in the same units as δ. The cash process X and the marked-to-market P&L, $\mathrm{PnL}_t = X_t + q_t S_t$, therefore satisfy

$$\mathrm{d}X_t = (\delta_t^+ - c)\,\Delta\,\mathrm{d}N_t^+ + (\delta_t^- - c)\,\Delta\,\mathrm{d}N_t^- - S_t\,\mathrm{d}q_t,$$

$$\mathrm{dPnL}_t = \mathrm{d}X_t + q_{t^-}\,\mathrm{d}S_t + S_t\,\mathrm{d}q_t = (\delta_t^+ - c)\,\Delta\,\mathrm{d}N_t^+ + (\delta_t^- - c)\,\Delta\,\mathrm{d}N_t^- + q_t\,\alpha_t\,\mathrm{d}t + q_t\,\sigma\,\mathrm{d}W_t,$$

where no covariation term appears because S is continuous.

2.4 Objective

Following CJR and Guéant [10], the market maker maximises the discounted expected P&L penalised by a running inventory cost,

$$\sup_{(\delta^+,\delta^-)\in\mathcal{A}} \mathbb{E}\left[\int_0^\infty e^{-rt}\Big(\mathrm{dPnL}_t - \psi(q_t)\,\mathrm{d}t\Big)\right], \qquad \psi(q) = \tfrac{1}{2}\gamma\sigma^2 q^2, \tag{1}$$

where $r > 0$ is a discount rate, $\gamma \geq 0$ a risk-aversion coefficient, and $\mathcal{A}$ the set of predictable processes valued in $\mathbb{R}_+^2$ (a post-only quote cannot cross the mid). The penalty is proportional to the variance of the inventory value. Since q is bounded, the stochastic integral $\int e^{-rt} q_t \sigma\,\mathrm{d}W_t$ has zero expectation, and (1) equals

$$\sup_{\mathcal{A}} \mathbb{E}\left[\int_0^\infty e^{-rt}\Big((\delta_t^+ - c)\Delta\,\mathrm{d}N_t^+ + (\delta_t^- - c)\Delta\,\mathrm{d}N_t^- + \big(\alpha_t q_t - \psi(q_t)\big)\,\mathrm{d}t\Big)\right]. \tag{2}$$

As in [12], we look for closed-loop controls (policies, in RL terms) $\delta_t^\pm = \delta^\pm(Y_{t^-})$, where Y is the state defined below. In the Poisson case Y reduces to the inventory; here it must also contain the order flow.

3 Dynamic Programming

3.1 State and jumps

The problem (2) is Markovian in the state

$$Y_t = (q_t, x_t), \qquad x_t = \big(\alpha_t,\ (R_{i,j}^{(p)}(t))_{i,j,p},\ \kappa_t^+,\ \kappa_t^-\big) \in \mathcal{X},$$

where x is the *market state*. The price S is not needed, since it only enters through α and the martingale term removed above. The horizon is infinite and the dynamics are time-homogeneous, so the value function does not depend on time:

$$V(y) = \sup_{\mathcal{A}} \mathbb{E}_y\left[\int_0^\infty e^{-rt}\Big((\delta_t^+ - c)\Delta\,\mathrm{d}N_t^+ + (\delta_t^- - c)\Delta\,\mathrm{d}N_t^- + \big(\alpha_t q_t - \psi(q_t)\big)\,\mathrm{d}t\Big)\right].$$

Between market orders. The state evolves continuously: q is constant, $\mathrm{d}\alpha_t = -\zeta\alpha_t\,\mathrm{d}t + \sigma_\alpha\,\mathrm{d}B_t$, $\mathrm{d}R_{i,j}^{(p)} = -\beta_{ij}^{(p)} R_{i,j}^{(p)}\,\mathrm{d}t$ and $\mathrm{d}\kappa_t^\pm = \beta_\kappa(\theta_\kappa - \kappa_t^\pm)\,\mathrm{d}t$. The associated generator, acting on a function $V(q,x)$ regular enough, is

$$\mathcal{L}V = -\zeta\alpha\,\partial_\alpha V + \tfrac{1}{2}\sigma_\alpha^2\,\partial^2_{\alpha\alpha} V - \sum_{i,j,p} \beta_{ij}^{(p)} R_{i,j}^{(p)}\,\partial_{R_{i,j}^{(p)}} V + \beta_\kappa \sum_{\pm} (\theta_\kappa - \kappa^\pm)\,\partial_{\kappa^\pm} V.$$

At a market order. Every market order moves the market state, filled or not; only the inventory differs (Table 1). We denote by T^+x (resp. T^-x) the market state after a buy (resp. sell) market order:

$$T^{\pm}x = \Big(\alpha \pm \epsilon^{\pm},\ R^{(p)}_{i,\pm} + 1\ \forall i,p,\ \ \kappa^{\pm} + \eta_\kappa,\ \ \kappa^{\mp} + \nu_\kappa\Big),$$

the other components being unchanged.

Table 1: Jumps of the state at a market order. A buy market order can only fill our ask (we sell, q decreases), a sell market order only our bid. $R^{(p)}_{i,j}$: target i, source j; all components p jump.

Event	Δq	$\Delta\alpha$	$\Delta R^{(p)}_{i,+}$	$\Delta R^{(p)}_{i,-}$	$\Delta\kappa^+$	$\Delta\kappa^-$
Buy MO, our ask filled	$-\Delta$	$+\epsilon^+$	$+1$	0	$+\eta_\kappa$	$+\nu_\kappa$
Buy MO, ask not filled	0	$+\epsilon^+$	$+1$	0	$+\eta_\kappa$	$+\nu_\kappa$
Sell MO, our bid filled	$+\Delta$	$-\epsilon^-$	0	$+1$	$+\nu_\kappa$	$+\eta_\kappa$
Sell MO, bid not filled	0	$-\epsilon^-$	0	$+1$	$+\nu_\kappa$	$+\eta_\kappa$

For compactness, we write $\widetilde{h}_\pm(\delta;q,\kappa) := h_\pm(\delta;\kappa)\,\mathbb{1}_{\{q\mp\Delta\in\mathcal{Q}\}}$ for the fill probability including the inventory constraint ($q-\Delta\in\mathcal{Q}$ iff $q>-Q$).

3.2 Hamilton–Jacobi–Bellman equation

Proposition 3.1 (HJB equation)**.** *Assume that V is C^2 in α and C^1 in (R,κ), with polynomial growth. Then, for all $y=(q,x)$,*

$$\begin{aligned} 0 = &-rV(q,x) + \mathcal{L}V(q,x) + \alpha q - \psi(q) \\ &+ \sum_{\pm} \lambda^{\pm}(x)\left[\sup_{\delta\geq 0}\widetilde{h}_\pm(\delta;q,\kappa^\pm)\Big((\delta-c)\Delta + \mathbb{A}^\pm V(q,x)\Big) + \mathbb{B}^\pm V(q,x)\right], \end{aligned} \tag{3}$$

where $\lambda^\pm(x)$ is the intensity in state x and

$$\mathbb{A}^\pm V(q,x) := V(q\mp\Delta, T^\pm x) - V(q,T^\pm x), \qquad \mathbb{B}^\pm V(q,x) := V(q,T^\pm x) - V(q,x).$$

$\mathbb{A}^\pm V$ is the value of an execution given that a market order of side $\pm$ has arrived, and $\mathbb{B}^\pm V$ the effect of a market order that does not fill us. The proof (Section A.1) applies the dynamic programming principle on $[0,h]$ and Itô's formula with jumps to $e^{-rt}V(Y_t)$, replaces the jump measures by their compensators, divides by h and lets $h\to 0^+$.

Remark 3.2. When $\epsilon^\pm = \eta_\kappa = \nu_\kappa = 0$ and $a^{(p)}_{ij} = 0$ (Poisson order flow, no drift), $T^\pm x = x$, $\mathbb{B}^\pm V = 0$, the state reduces to q and (3) becomes the stationary equation of Guéant [10]. With a self-exciting flow, a market order that does not reach our quote still changes the value through the intensities and the drift.

3.3 Optimal quotes

The supremum in (3) is a one-dimensional problem for each side.

Lemma 3.3 (Optimal quote)**.** *Let h satisfy Assumption 2.3, $k\in\mathbb{R}$, and $F(\delta) := h(\delta)\,(\delta+k)$ on $\mathbb{R}_+$. Define $g(\delta) := \delta + k + h(\delta)/h'(\delta)$. Then g is strictly increasing, and F admits a unique maximiser on $\mathbb{R}_+$:*

$$\delta^* = 0 \ \ \textit{if } g(0)\geq 0, \qquad \textit{otherwise} \qquad \delta^* = \tilde{\delta},\ \textit{the unique root of}\ \ \delta = -k - \frac{h(\delta)}{h'(\delta)}.$$

Proof. We have $F'(\delta) = h'(\delta)(\delta + k) + h(\delta) = h'(\delta)\, g(\delta)$, and

$$g'(\delta) = 1 + \frac{h'(\delta)^2 - h(\delta)h''(\delta)}{h'(\delta)^2} = 2 - \frac{h(\delta)h''(\delta)}{h'(\delta)^2} \;\geq\; \varepsilon := 2 - \sup_{\delta' \geq 0} \frac{hh''}{(h')^2} > 0$$

by (H4). Hence g is strictly increasing and $g(\delta) \geq g(0) + \varepsilon\delta \to +\infty$. Since $h' < 0$ by (H2), F' has the sign of $-g$. If $g(0) \geq 0$, F is non-increasing on $\mathbb{R}_+$ and $\delta^* = 0$. Otherwise g has a unique root $\tilde{\delta} > 0$, F increases on $[0, \tilde{\delta}]$ and decreases afterwards (it is strictly quasi-concave), so $\tilde{\delta}$ is the unique maximiser. The root satisfies $\tilde{\delta} + k + h(\tilde{\delta})/h'(\tilde{\delta}) = 0$. □

Corollary 3.4 (Optimal quotes in feedback form)**.** *The supremum in* (3) *is attained at* $\delta^{\pm,*}(q,x)$ *given by Lemma 3.3 with* $h = h_\pm(\cdot\,;\kappa^\pm)$ *and* $k = \mathbb{A}^\pm V(q,x)/\Delta - c$*. In the exponential case* $h_\pm(\delta;\kappa) = e^{-\kappa\delta}$*,* $-h/h' = 1/\kappa$ *and*

$$\boxed{\delta^{\pm,*}(q,x) = \left(\frac{1}{\kappa^\pm} + c - \frac{V(q \mp \Delta, T^\pm x) - V(q, T^\pm x)}{\Delta}\right)^+} \tag{4}$$

and the maximum value is, when the positive part is not active,

$$\sup_{\delta \geq 0} e^{-\kappa^\pm \delta}\Big((\delta - c)\Delta + \mathbb{A}^\pm V\Big) = \frac{\Delta}{\kappa^\pm} \exp\Big(-1 - \kappa^\pm c + \kappa^\pm\, \mathbb{A}^\pm V/\Delta\Big).$$

Proof. Divide the bracket by $\Delta > 0$ and apply Lemma 3.3 with $k = \mathbb{A}^\pm V/\Delta - c$. For the exponential case, $\tilde{\delta} = 1/\kappa - k$ and $g(0) \geq 0$ iff $\tilde{\delta} \leq 0$, hence the positive part. Plugging δ^* into $e^{-\kappa\delta}(\delta + k)\Delta$ gives $(\Delta/\kappa)e^{-\kappa\tilde{\delta}} = (\Delta/\kappa)e^{-1+\kappa k}$. □

As in CJR, the quote is $1/\kappa + c$ minus the marginal value of the inventory change, evaluated after the market has reacted to the order ($T^\pm x$). This is where adverse selection enters: after a buy order, α and the buy intensity increase, being short is worth less, and the ask widens.

4 Event-Time Formulation

Equation (3) has $1+1+4P+2 = 10$ dimensions. Since the control only matters at the arrival of market orders, we rewrite it, following [12], as a Bellman equation between consecutive market orders. We assume $\sigma_\alpha = 0$ (Remark 2.5); see Remark 4.6 for $\sigma_\alpha > 0$.

4.1 Deterministic flow and waiting time

With $\sigma_\alpha = 0$, the state evolves deterministically between two market orders. Let $\Phi_u(q,x) = (q, \phi_u(x))$ denote this flow,

$$\phi_u(x) = \Big(\alpha e^{-\zeta u},\ (R^{(p)}_{i,j} e^{-\beta^{(p)}_{ij} u})_{i,j,p},\ \theta_\kappa + (\kappa^\pm - \theta_\kappa)e^{-\beta_\kappa u}\Big).$$

Let τ be the time until the next market order and $\bar{\lambda}(x) := \lambda^+(x) + \lambda^-(x)$ the total intensity.

Proposition 4.1 (Survival function of the waiting time)**.** *Starting from the market state* x*,*

$$\begin{aligned}\mathcal{S}_x(u) := \mathbb{P}(\tau > u \mid x) &= \exp\left(-\int_0^u \bar{\lambda}(\phi_v(x))\,\mathrm{d}v\right) \\ &= \exp\left(-(\mu_+ + \mu_-)\,u - \sum_{i,j\in\{+,-\}} \sum_{p=1}^{P} \frac{a^{(p)}_{ij}}{\beta^{(p)}_{ij}} R^{(p)}_{i,j}\big(1 - e^{-\beta^{(p)}_{ij} u}\big)\right),\end{aligned}$$

and τ *has density* $f_x(u) = \bar{\lambda}(\phi_u(x))\,\mathcal{S}_x(u)$*. In particular,* $\mathcal{S}_x$ *does not depend on the quotes.*

Proof. The first equality is the compensator/survival identity of [28, Prop. 5.2], applied to the superposed process $M^+ + M^-$, whose intensity is $\bar\lambda$. Between events $\lambda^i(\phi_v(x)) = \mu_i + \sum_{j,p} a_{ij}^{(p)} R_{i,j}^{(p)} e^{-\beta_{ij}^{(p)} v}$; integrating on $[0,u]$ gives the closed form. The quotes affect which orders fill us, not when orders arrive. □

This extends the univariate formula used for exact simulation in [28].

Lemma 4.2 (Expected discount and running terms)**.** *For all* (q,x)*,*

(i) $\mathbb{E}[e^{-r\tau} \mid x] = \displaystyle\int_0^\infty e^{-ru}\,\bar\lambda(\phi_u(x))\,\mathcal{S}_x(u)\,\mathrm{d}u = 1 - r\int_0^\infty e^{-ru}\mathcal{S}_x(u)\,\mathrm{d}u;$

(ii) $\mathbb{E}\left[\displaystyle\int_0^\tau e^{-ru}\psi(q)\,\mathrm{d}u \;\Big|\; x\right] = \psi(q)\int_0^\infty e^{-ru}\mathcal{S}_x(u)\,\mathrm{d}u;$

(iii) $\mathbb{E}\left[\displaystyle\int_0^\tau e^{-ru}\alpha_u q\,\mathrm{d}u \;\Big|\; x\right] = q\,\alpha\int_0^\infty e^{-(r+\zeta)u}\mathcal{S}_x(u)\,\mathrm{d}u.$

Proof. (i) The first equality uses the density of τ. For the second, integrate by parts with $\frac{\mathrm{d}}{\mathrm{d}u}\mathcal{S}_x(u) = -\bar\lambda(\phi_u(x))\mathcal{S}_x(u)$: $\int_0^\infty e^{-ru}(-\mathcal{S}_x'(u))\,\mathrm{d}u = \big[-e^{-ru}\mathcal{S}_x(u)\big]_0^\infty - r\int_0^\infty e^{-ru}\mathcal{S}_x(u)\,\mathrm{d}u$, and the bracket equals 1 since $\mathcal{S}_x(0)=1$ and $\mathcal{S}_x \le 1$. (ii) Write $\int_0^\tau = \int_0^\infty \mathbb{1}_{\{u<\tau\}}$; the integrand is non-negative, so Fubini–Tonelli gives $\psi(q)\int_0^\infty e^{-ru}\mathbb{P}(u<\tau \mid x)\,\mathrm{d}u$. (iii) On $\{u<\tau\}$ no market order has arrived, so $\alpha_u = \alpha e^{-\zeta u}$, which is deterministic; Fubini applies because $\int_0^\infty e^{-ru}\,|q\alpha|\,e^{-\zeta u}\,\mathrm{d}u < \infty$. □

We gather the running terms in

$$C(q,x) := \mathbb{E}\left[\int_0^\tau e^{-ru}(\alpha_u q - \psi(q))\,\mathrm{d}u \;\Big|\; x\right] = \int_0^\infty e^{-ru}\,\mathcal{S}_x(u)\,(q\,\alpha e^{-\zeta u} - \psi(q))\,\mathrm{d}u.$$

4.2 Event-time Bellman equation

Let $\widetilde V(q,x)$ be the value between two events, (q,x) being the state just after the last market order, and $\Theta(q,z)$ the value at the arrival of a market order, z being the market state just before it and its side not yet known. $\widetilde V$ is expressed with Θ, and Θ with $\widetilde V$.

Proposition 4.3 (Event-time Bellman equation)**.** *Assume* $\sigma_\alpha = 0$*. Then*

$$\widetilde V(q,x) = C(q,x) + \int_0^\infty e^{-ru}\,\bar\lambda(\phi_u(x))\,\mathcal{S}_x(u)\,\Theta(q,\phi_u(x))\,\mathrm{d}u, \tag{5}$$

$$\Theta(q,z) = \sum_{\pm}\frac{\lambda^\pm(z)}{\bar\lambda(z)}\sup_{\delta\ge 0}\Big\{\widetilde h_\pm\Big((\delta-c)\Delta + \widetilde V(q\mp\Delta, T^\pm z)\Big) + (1-\widetilde h_\pm)\,\widetilde V(q,T^\pm z)\Big\}, \tag{6}$$

where $\widetilde h_\pm = \widetilde h_\pm(\delta; q, \kappa^\pm)$*,* $\kappa^\pm$ *being read in* z*.*

Proof. Equation (5)*.* Split the criterion at the first market order τ. On $[0,\tau)$ the inventory is constant and the expected running reward is $C(q,x)$ (Lemma 4.2). By the strong Markov property and time-homogeneity, the remaining value is $e^{-r\tau}\Theta(Y_{\tau^-})$. Since $\sigma_\alpha = 0$, $Y_{\tau^-} = \Phi_\tau(q,x)$, so this is $g(\tau)$ with $g(u) = e^{-ru}\Theta(q,\phi_u(x))$, whose expectation is $\int_0^\infty g(u) f_x(u)\,\mathrm{d}u$.

Equation (6)*.* Given that a market order arrives in state z, the probability that it is a buy is, by Bayes' formula on an interval $\mathrm{d}t$ (simultaneous arrivals have probability $o(\mathrm{d}t)$),

$$\frac{\lambda^+(z)\,\mathrm{d}t}{\lambda^+(z)\,\mathrm{d}t + \lambda^-(z)\,\mathrm{d}t} = \frac{\lambda^+(z)}{\bar\lambda(z)},$$

and $\lambda^-(z)/\bar\lambda(z)$ for a sell. A buy order fills our ask with probability $\widetilde h_+(\delta^+)$, in which case we earn $(\delta^+ - c)\Delta$ and move to $(q-\Delta, T^+z)$; otherwise we move to (q, T^+z). The value is then $\widetilde V$ at the new state. The quote only affects this fill, so it maximises the conditional expectation. The sell side is symmetric. □

The quote only appears in Θ: $\widetilde{V}$ averages over when the next order arrives, Θ over its side and its fill. Substituting (6) into (5), the total intensity cancels:

Corollary 4.4 (Closed equation for $\widetilde{V}$)**.**

$$\widetilde{V}(q,x) = C(q,x) + \int_0^\infty e^{-ru}\,\mathcal{S}_x(u) \sum_{\pm} \lambda^{\pm}(\phi_u x) \sup_{\delta\geq 0} \Big\{ \widetilde{h}_{\pm}\big((\delta - c)\Delta + \widetilde{V}(q \mp \Delta, T^{\pm}\phi_u x)\big) + (1 - \widetilde{h}_{\pm})\,\widetilde{V}(q, T^{\pm}\phi_u x) \Big\}\,\mathrm{d}u$$

where $\widetilde{h}_{\pm} = \widetilde{h}_{\pm}(\delta; q, \kappa^{\pm}(\phi_u x))$.

Remark 4.5 (Consistency with the HJB equation). The bracket of (3) can be written $\bar{\lambda}(\Theta - V)$:

$$\sum_{\pm} \lambda^{\pm} \Big[\sup_{\delta} \widetilde{h}_{\pm}\big((\delta - c)\Delta + \mathbb{A}^{\pm} V\big) + \mathbb{B}^{\pm} V \Big] = \bar{\lambda}(x)\,\big(\Theta(q,x) - V(q,x)\big),$$

so that (3) reads $(r + \bar{\lambda})V = \mathcal{L}V + \alpha q - \psi(q) + \bar{\lambda}\,\Theta$. With $\sigma_\alpha = 0$, $\mathcal{L}$ is the derivative along the flow and (5) is the solution of this equation along $u \mapsto \phi_u(x)$, with integrating factor $e^{-ru}\mathcal{S}_x(u)$. Hence $\widetilde{V} = V$.

Remark 4.6 (Case $\sigma_\alpha > 0$). Without jumps, α is an Ornstein–Uhlenbeck process: on $\{u < \tau\}$, $\alpha_u = \tilde{\alpha}(u) := \alpha e^{-\zeta u} + \sigma_\alpha \int_0^u e^{-\zeta(u-v)}\,\mathrm{d}B_v \sim \mathcal{N}\big(\alpha e^{-\zeta u}, \frac{\sigma_\alpha^2}{2\zeta}(1 - e^{-2\zeta u})\big)$ (Section A.2). The intensities do not depend on α, so τ is independent of B, and $\mathcal{S}_x$ and C are unchanged (Fubini applies by Cauchy–Schwarz and Itô's isometry). Only the pre-event state becomes random:

$$\widetilde{V}(q,x) = C(q,x) + \int_0^\infty e^{-ru}\bar{\lambda}(\phi_u x)\mathcal{S}_x(u)\,\mathbb{E}\Big[\Theta\big(q, \tilde{\alpha}(u), R(u), \kappa(u)\big)\Big]\,\mathrm{d}u,$$

Remark 4.7 (Comparison with Guéant–Manziuk). In [12] the state does not move between two requests, so $\mathbb{E}[e^{-r\tau}\Theta(q)] = \frac{\bar{\lambda}}{r+\bar{\lambda}}\Theta(q)$ and the critic can learn Θ directly. Here the state at the event depends on τ. Besides, to improve its quote in state z the actor needs the bracket of (6),

$$\mathcal{Q}_{\pm}(q,z,\delta) := \widetilde{h}_{\pm}(\delta)\Big((\delta - c)\Delta + \widetilde{V}(q \mp \Delta, T^{\pm}z)\Big) + \big(1 - \widetilde{h}_{\pm}(\delta)\big)\widetilde{V}(q, T^{\pm}z),$$

which involves $\widetilde{V}$, not Θ. Recovering $\widetilde{V}$ from Θ requires the integral along the flow, which has no closed form here. We therefore learn the value at post-decision states [25], $\widetilde{V}$, or directly the action-value $\mathcal{Q}$ (Section 5).

4.3 Contraction and sampled targets

Let $\mathcal{T}$ denote the right-hand side of Corollary 4.4, as an operator on bounded functions W of (q,x).

Assumption 4.8 (Bounded intensity)**.** The market state takes values in a set $\mathcal{X}_b$ on which $\bar{\lambda} \leq \Lambda_{\max} < \infty$ and $|\alpha| \leq \alpha_{\max}$.

The truncation is needed because $\bar{\lambda}$ is unbounded; it holds in any simulation of finite length.

Proposition 4.9 (Contraction)**.** *For bounded* W_1, W_2*,*

$$|\mathcal{T}W_1 - \mathcal{T}W_2|\,(q,x) \leq \mathbb{E}\big[e^{-r\tau} \mid x\big]\ \|W_1 - W_2\|_\infty\,.$$

Under Assumption 4.8, $\mathbb{E}[e^{-r\tau} \mid x] \leq \frac{\Lambda_{\max}}{r+\Lambda_{\max}} < 1$*, so* $\mathcal{T}$ *is a contraction for* $\|\cdot\|_\infty$ *and* $\widetilde{V}$ *is its unique bounded fixed point.*

See Section A.3 for the proof. With $r = 1/60\ \mathrm{s}^{-1}$ and about 3.8 orders per second (Table 3), the modulus is close to $\bar{\lambda}/(r + \bar{\lambda}) \approx 0.996$, an effective horizon of about 230 events (one minute).

Sampled target. Let (q, x) be a post-event state. Simulate the next market order: its time τ from $\mathcal{S}_x$, the pre-event state $z = \phi_\tau(x)$, its side $s \in \{+, -\}$ with probability $\lambda^s(z)/\bar{\lambda}(z)$, and a fill $F \sim \text{Bernoulli}(\widetilde{h}_s(\delta^s))$. Write (q', x') for the next post-event state. Then

$$G := \frac{q\,\alpha}{r+\zeta}(1 - e^{-(r+\zeta)\tau}) - \frac{\psi(q)}{r}(1 - e^{-r\tau}) + e^{-r\tau}\Big(F\,(\delta^s - c)\Delta + W(q', x')\Big), \tag{7}$$

where the first two terms are $\int_0^\tau e^{-ru}(q\,\alpha e^{-\zeta u} - \psi(q))\,\mathrm{d}u$. G is an unbiased estimate of $\mathcal{T}^\delta W(q, x)$, the Bellman operator for a fixed quote δ, and $G - W(q, x)$ is the temporal difference error minimised by the critics. The only noise comes from (τ, s, F); the fill can also be integrated exactly by using $\mathcal{Q}_s$.

4.4 The Poisson special case

Corollary 4.10 (Poisson order flow)**.** *Let $a_{ij}^{(p)} = 0$ (so $\lambda^\pm = \mu_\pm$, $\bar{\lambda} = \mu_+ + \mu_-$), $\eta_\kappa = \nu_\kappa = 0$, $\alpha \equiv 0$ and $h_\pm(\delta) = e^{-\kappa\delta}$. The state reduces to q and*

$$\widetilde{V}(q) = \frac{-\psi(q)}{r+\bar{\lambda}} + \frac{\bar{\lambda}}{r+\bar{\lambda}} \sum_\pm \frac{\mu_\pm}{\bar{\lambda}} \sup_{\delta \geq 0} \Big\{\widetilde{h}_\pm(\delta)((\delta - c)\Delta + \widetilde{V}(q \mp \Delta)) + (1 - \widetilde{h}_\pm(\delta))\widetilde{V}(q)\Big\}.$$

With the optimal quote of Corollary 3.4, when the positive part is inactive, this is equivalent to

$$r\,\widetilde{V}(q) = -\psi(q) + \sum_\pm \mu_\pm\, \mathbb{1}_{\{q \mp \Delta \in \mathcal{Q}\}}\, \frac{\Delta}{\kappa} \exp\left(-1 - \kappa c + \kappa\, \frac{\widetilde{V}(q \mp \Delta) - \widetilde{V}(q)}{\Delta}\right).$$

Proof. Here $\mathcal{S}_x(u) = e^{-\bar{\lambda}u}$, so $C(q) = -\psi(q)/(r + \bar{\lambda})$ and $\int_0^\infty e^{-ru}\bar{\lambda}e^{-\bar{\lambda}u}\,\mathrm{d}u = \bar{\lambda}/(r + \bar{\lambda})$ in (5); Θ follows from (6) with $T^\pm z = z$. For the second form, the supremum equals $\widetilde{V}(q) + \frac{\Delta}{\kappa} e^{-1-\kappa c + \kappa(\widetilde{V}(q\mp\Delta) - \widetilde{V}(q))/\Delta}$ by Corollary 3.4; multiply by $r + \bar{\lambda}$ and simplify the $\bar{\lambda}\widetilde{V}(q)$ terms. □

This is the discounted version of the equations of [11, 10], a system of $2Q/\Delta + 1$ equations solved by value iteration (modulus $\bar{\lambda}/(r + \bar{\lambda})$). We use it to test the critics before the Hawkes case. Figure 1 shows the quotes obtained with the calibrated mean intensities.

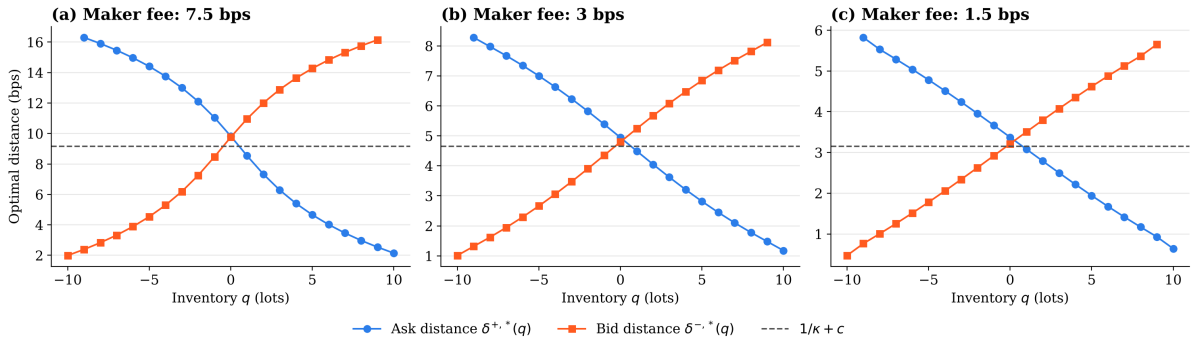


Figure 1: Optimal quotes of the Poisson special case (Corollary 4.10), obtained by value iteration, with the stationary mean intensities and the fill parameter of Table 3, $\gamma = 0.01$, $r = 1/60$ s^{-1}, $Q = 10$ lots.

5 Deep Reinforcement Learning

5.1 Actor and critic

An actor–critic method [18, 29] learns two neural networks. The *actor* π_ϕ is the policy: it maps a state to an action, here the two quotes. The *critic* estimates the value of a state, or

of a state–action pair, under the current policy, by minimising a temporal difference error such as (7). The actor is then moved in the direction that increases the critic, and the two steps alternate (approximate policy iteration). Here the critic plays the role of $\widetilde{V}$ or $\mathcal{Q}$, and the actor that of Lemma 3.3.

5.2 Semi-Markov decision process

Let τ_n be the arrival time of the n-th market order and $Z_n := Y_{\tau_n^-}$ the pre-event state (the state z of Θ). Decisions are taken at these times, which gives a semi-Markov decision process (SMDP) [3, 26]:

- **State.** $Z_n = (q, \alpha, R, \kappa)$, of dimension $2+2P = 6$ in the implemented model (Remark 2.5). The network inputs are q/Q, α/s_α (with s_α the standard deviation of α after a burn-in) and $\log(1 + R_j^{(p)})$.
- **Action.** $A_n = (\kappa\delta^+, \kappa\delta^-) \in \mathbb{R}_+^2$, minus the log of the fill probabilities; a side blocked by the inventory bound is masked.
- **Transition and reward.** The side s is drawn with probability $\lambda^s(Z_n)/\bar{\lambda}(Z_n)$; the fill with probability $e^{-A_n^s}$; the state jumps as in Table 1; then the next market order arrives after τ, drawn by thinning [22], and the state follows the flow. The reward is

$$\mathcal{R}_n = F_n\,(\delta_n^s - c)\Delta + \frac{q'\alpha'}{r+\zeta}(1 - e^{-(r+\zeta)\tau}) - \frac{\psi(q')}{r}(1 - e^{-r\tau}),$$

  where (q', α') is the post-event state. The Brownian term of the price is integrated out in expectation.
- **Discount.** $D_n := e^{-r(\tau_{n+1}-\tau_n)}$. It is random: close to one during bursts, smaller in calm periods.

The critic learns the action-value at pre-event states,

$$\mathcal{Q}^\pi(z, a) = \mathbb{E}\big[\mathcal{R}_n + D_n\,\mathcal{Q}^\pi(Z_{n+1}, \pi(Z_{n+1}))\,\big|\,Z_n = z, A_n = a\big],$$

whose optimum relates to Section 4 by

$$\mathcal{Q}^*(z, a) = \sum_{\pm} \frac{\lambda^\pm(z)}{\bar{\lambda}(z)}\,\mathcal{Q}_\pm(q, z, a^\pm/\kappa), \qquad \Theta(q, z) = \sup_{a \in \mathbb{R}_+^2} \mathcal{Q}^*(z, a),$$

since $\widetilde{V}$ already contains the running reward until the next order. The optimal actor is the quote of Corollary 3.4 at the pre-event state. TD3 and SAC do not use h, λ or $T^\pm$, only sampled transitions.

Architecture. Actor and critics are multilayer perceptrons with two hidden layers of 64 units and SiLU activations [7]. The actor outputs $A = \mathrm{softplus}(\cdot) \geq 0$, initialised at $\delta^\pm = 1/\kappa + c$. Each critic takes (α, R) and $\log(1 + A)$ as inputs and has one output per inventory level, the output of the current q being selected. Rewards are divided by $\frac{1}{e\kappa}\frac{\bar{\lambda}_\infty}{r}$, the value of quoting at $1/\kappa$ forever, with $\bar{\lambda}_\infty = \mathbf{1}^\top(I - \Gamma)^{-1}\boldsymbol{\mu}$ the stationary total intensity.

Remark 5.1 (Model-based variant). We also implemented a model-based actor–critic close to [12], whose critic learns $\widetilde{V}$ with the target (7) (fill integrated exactly) and whose quotes are given by (4). It was used to check the event-time formulation; the experiments use TD3 and SAC.

5.3 TD3

TD3 [9] extends DDPG [20, 27]. The actor is deterministic, $a = \pi_\phi(z)$, and two critics $\mathcal{Q}_{w_1}, \mathcal{Q}_{w_2}$ are learned with slowly updated target copies [21]. The target takes the minimum of the two target critics, which limits overestimation; noise is added to the target action (target policy smoothing); and the actor is updated once every d critic updates. In our SMDP the constant discount factor is replaced by D_n (Algorithm 1).

Algorithm 1 TD3 in the event-time SMDP (one iteration)

Input: actor π_ϕ, critics $\mathcal{Q}_{w_1}, \mathcal{Q}_{w_2}$, targets $\pi_{\bar\phi}, \mathcal{Q}_{\bar w_1}, \mathcal{Q}_{\bar w_2}$, replay buffer $\mathcal{D}$
$a \leftarrow \big(\pi_\phi(z) + \varepsilon\big)^+, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2_{\text{expl}} I)$ ▷ exploration
Simulate the next market order: $(\mathcal{R}, D, z')$; store $(z, a, \mathcal{R}, D, z')$ in $\mathcal{D}$
Sample a minibatch $\{(z_k, a_k, \mathcal{R}_k, D_k, z'_k)\}$ from $\mathcal{D}$
$a'_k \leftarrow \big(\pi_{\bar\phi}(z'_k) + \text{clip}(\varepsilon'_k, -c_\varepsilon, c_\varepsilon)\big)^+, \ \varepsilon'_k \sim \mathcal{N}(0, \sigma^2_{\text{targ}} I)$ ▷ smoothing
$G_k \leftarrow \mathcal{R}_k + D_k \min_{i=1,2} \mathcal{Q}_{\bar w_i}(z'_k, a'_k)$ ▷ clipped double Q, random discount
Update w_i by a gradient step on $\sum_k \big(\mathcal{Q}_{w_i}(z_k, a_k) - G_k\big)^2$, $i = 1, 2$
if iteration $\equiv 0 \bmod d$ **then** ▷ delayed updates
    Update ϕ by a gradient step on $-\sum_k \mathcal{Q}_{w_1}(z_k, \pi_\phi(z_k))$
    $\bar w_i \leftarrow (1-\varrho)\bar w_i + \varrho w_i, \quad \bar\phi \leftarrow (1-\varrho)\bar\phi + \varrho\phi$ ▷ Polyak averaging
end if

5.4 SAC

SAC [13, 14] maximises the reward plus the entropy of a stochastic policy. The actor outputs a Gaussian $\xi \sim \mathcal{N}(m_\phi(z), s_\phi(z)^2)$ per side, squashed as $a = \text{softplus}(\xi)$; the log-density of a includes the Jacobian $\log \sigma(\xi)$ of the softplus. The soft target is

$$G = \mathcal{R} + D\Big(\min_i \mathcal{Q}_{\bar w_i}(z', a') - \vartheta \log \pi_\phi(a' \mid z')\Big), \qquad a' \sim \pi_\phi(\cdot \mid z'),$$

and the actor minimises $\mathbb{E}\big[\vartheta \log \pi_\phi(a \mid z) - \min_i \mathcal{Q}_{w_i}(z, a)\big]$. The temperature ϑ is tuned so that the entropy tracks minus the number of active sides. At evaluation, $a = \text{softplus}(m_\phi(z))$.

5.5 Training pipeline

Synthetic training. TD3 and SAC are trained in the calibrated simulator (64 parallel environments, up to 8.4 million transitions), with one run per algorithm, fee level and seed (41, 53, 67). Checkpoints are scored every 262,144 transitions on fixed simulated paths, with early stopping after 8 evaluations without improvement.

Fine-tuning. The best checkpoints are then fine-tuned, still on simulated transitions; historical data only enter through selection. The fine-tuning starts from the checkpoint with lower learning rates and a critic warm-up, and adds an anchor to the initial actor π_{ϕ_0}, as in TD3+BC [8]:

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_{\text{RL}} + w_n \,\overline{|\mathcal{Q}|}\, \mathbb{E}\big[\,\|\pi_\phi(z) - \pi_{\phi_0}(z)\|^2\,\big], \qquad w_n = w_0\Big(1 - \frac{n}{N_{\text{anchor}}}\Big)^+,$$

An exponential moving average of the actor is also evaluated, and some runs use domain randomisation [30] of μ, a, $\epsilon^\pm$ and ζ.

Selection. Every 131,072 transitions, the actors are replayed on the 12 validation sessions of 2024. Hyperparameters (Table 5) are chosen on seeds 41 and 53; then, for each seed, we keep the checkpoint with the best mean daily net P&L among those with at least 5 fills per day and a mean absolute inventory below 5 lots. This gives 18 policies.

6 Empirical Results

6.1 Data and calibration

We use Binance spot DOTUSDT trades and best quotes (tick 0.001 USDT) from Tardis, on the first day of each month, 00:10–23:59 UTC. Market orders are trades with the same timestamps and side; their depth is the distance between their extreme price and the previous mid. Prices are in bps of the mid at 00:10, and a lot is 1,000 USDT.

Table 2: Splits of the monthly sessions.

Split	Period	Sessions	Use
Training	Jan. 2021 – Dec. 2023	36	Calibration of the simulator
Validation	Jan. 2024 – Dec. 2024	12	Selection of hyperparameters and checkpoints
Test	Jan. 2025 – Oct. 2026	22	Final historical replay

The parameters are calibrated on the training sessions (Table 3).

- **Hawkes.** The decays are fixed at $\beta^{(1)} = 2$ and $\beta^{(2)} = 0.05$ s^{-1}. The log-likelihood, computed with the recursion of [28, Prop. 4.4], is then concave in (μ, a) and maximised by L-BFGS-B [5]. We check that $\rho(\Gamma) < 1$.
- **Alpha.** For each ζ in a grid, $\epsilon^{\pm}$ are estimated by non-negative least squares [19] of the 10-second mid change on its conditional expectation; we keep the ζ with the best R^2.
- **Volatility.** σ^2 is the variance of 60-second mid increments net of the drift.
- **Fill probability.** κ is fitted on $\log \mathbb{P}(\text{depth} \geq \delta)$ by least squares through the origin (Figure 2).

Table 3: Calibrated parameters (DOTUSDT). Intensities in s^{-1}, δ and $\epsilon^{\pm}$ in bps, σ^2 in bps^2 s^{-1}. $a_{ij}^{(p)}$: target i, source j.

Parameter	Value	Parameter	Value
μ_+	0.0890	μ_-	0.0756
$a_{++}^{(1)}$	1.6860	$a_{--}^{(1)}$	1.7067
$a_{+-}^{(1)}$	0.0570	$a_{-+}^{(1)}$	0.0426
$a_{++}^{(2)}$	0.0012	$a_{--}^{(2)}$	0.0019
$a_{+-}^{(2)}$	0.0035	$a_{-+}^{(2)}$	0.0019
$\eta = \rho(\Gamma)$	0.957	$\bar{\lambda}_\infty^+,\ \bar{\lambda}_\infty^-$	2.020, 1.801
ζ	1.0	$\epsilon^+,\ \epsilon^-$	0.0017, 0
σ^2	3.246	κ	0.607
γ	0.01	r	1/60
Q	10 lots	Δ	1 lot

The flow is close to critical ($\eta = 0.96$). Self-excitation sits in the fast component and cross-excitation is about thirty times weaker. The drift explains little ($R^2 < 10^{-3}$), and the impact of sell orders is estimated at zero. Finally, $1/\kappa \approx 1.6$ bps: the median depth of market orders is about 1.2 bps and the 99th percentile about 8.6 bps. With a flat inventory the optimal quote is about $1/\kappa + c$, so at $c = 7.5$ bps it lies where fewer than 1% of market orders reach. Figure 3 shows a simulated path.

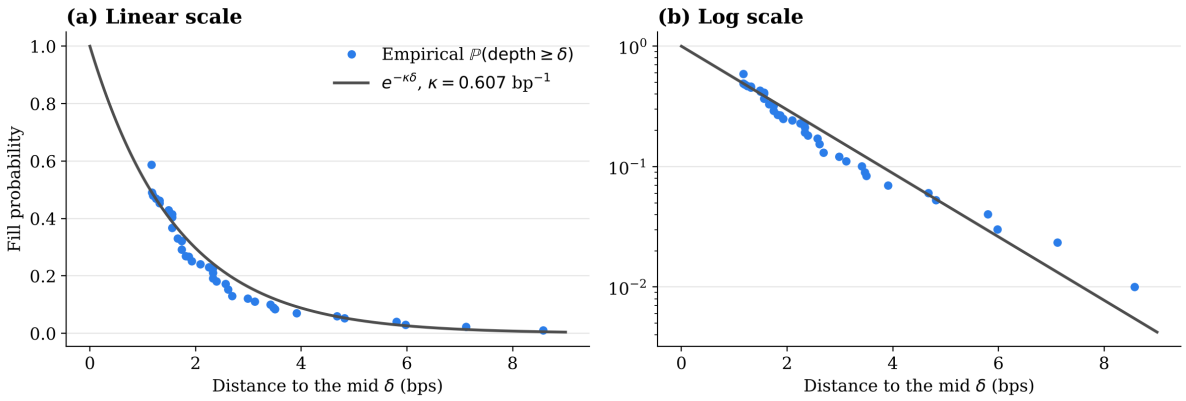


Figure 2: Empirical probability that a market order reaches depth δ, and the fitted exponential fill probability $h(\delta) = e^{-\kappa\delta}$.

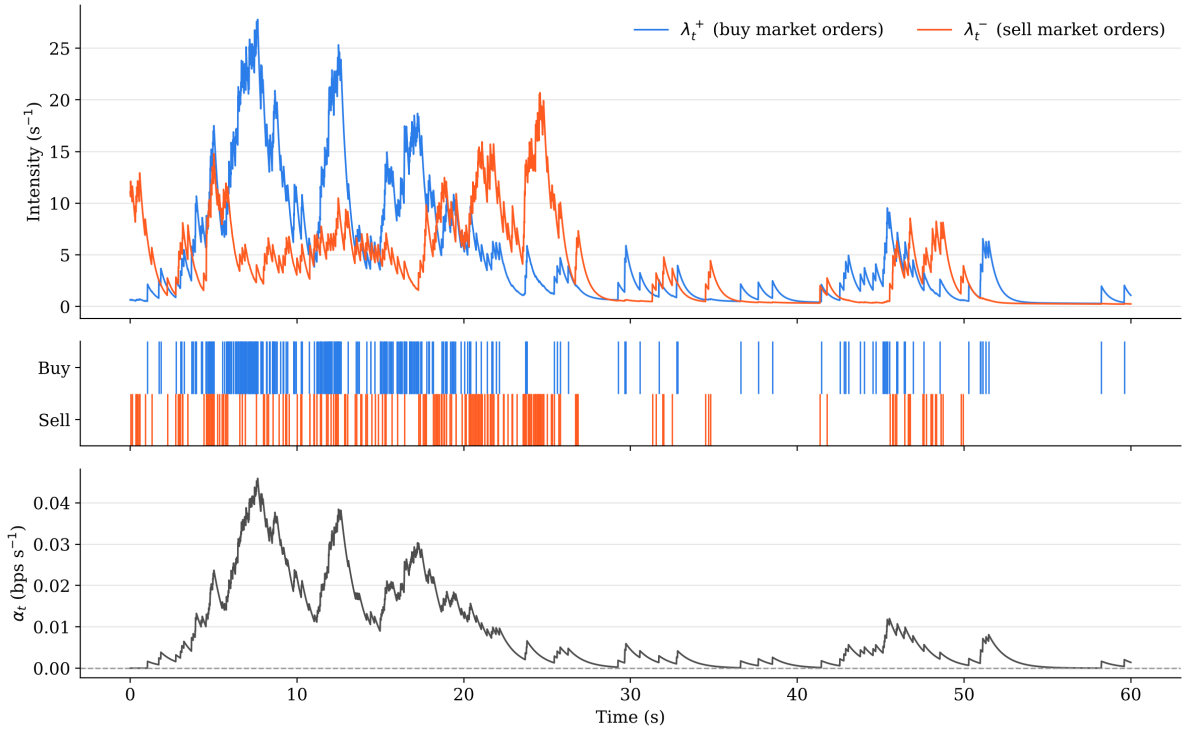


Figure 3: One minute of the calibrated model: intensities $\lambda^{\pm}$, market orders, and drift α, simulated by thinning after a burn-in.

6.2 Historical replay

The policies are replayed causally on the observed sessions. Decisions are taken every 0.1 s, from the market orders received strictly before. Quotes are rounded to the tick away from the mid, posted after a 0.1 s latency, and post-only. Under the m-tick rule, a quote is filled when the extreme price of a market order of the right side goes beyond it by at least m ticks, at most once per side and decision. The primary rule is $m = 1$, which partly accounts for queue priority; $m = 0$ and $m = 2$ are sensitivity checks. Maker fees are charged on each fill and the inventory is liquidated at 23:59 with a taker fee. Our orders have no market impact. We use three Binance fee levels (maker/taker, bps): 7.5/7.5, 3/4.5 and 1.5/2.175. The benchmark is

Avellaneda–Stoikov [1] with a rolling horizon $H = 60$ s,

$$\delta^{\pm}_{\mathrm{AS}}(q) = \left(\tfrac{1}{\gamma}\log\left(1+\tfrac{\gamma}{\kappa}\right) + \tfrac{1}{2}\gamma\sigma^2 H + c \mp q\,\gamma\sigma^2 H\right)^{+},$$

with the same units and inventory bounds.

6.3 Results

Table 4 and Figure 4 give the mean net P&L per test session under the one-tick rule. All three methods lose money. SAC is best at 3 bps, TD3 at 1.5 bps and AS at 7.5 bps. The dispersion between seeds is often larger than the gap between methods: at 1.5 bps, one TD3 seed earns about 33 USDT per session and another loses about 49.

Table 4: Mean net P&L per test session (USDT), one-tick rule, 22 sessions. TD3 and SAC are averaged over three training seeds. The last two columns are paired differences from AS (positive values favour the learned policy).

	Mean net P&L			Difference from AS	
Maker / taker fee (bps)	TD3	SAC	AS	TD3	SAC
7.5 / 7.5	−37.85	−25.84	**−21.83**	−16.02	−4.01
3 / 4.5	−31.46	**−19.56**	−26.19	−5.27	+6.63
1.5 / 2.175	**−12.42**	−26.65	−26.52	+14.10	−0.13

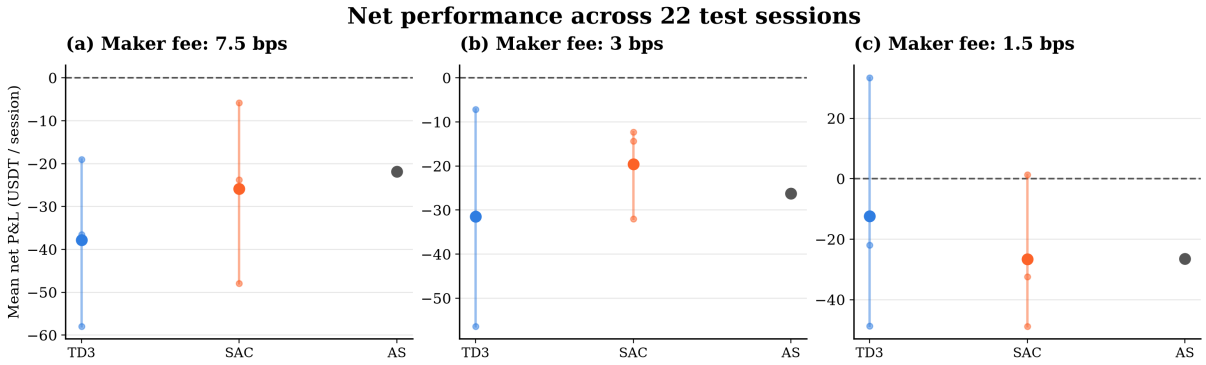


Figure 4: Mean net P&L across the 22 test sessions, one-tick rule. Large dots: means over the three training seeds; small dots and segments: individual seeds and their range (not confidence intervals). Panel scales differ.

Figure 5 shows the paired differences with AS, with the 2.5–97.5% quantiles of a circular block bootstrap [24] (blocks of two sessions, 10,000 draws, seeds averaged first). TD3 at 1.5 bps gains about 14 USDT per session and SAC at 3 bps about 7, but all intervals contain zero. These intervals ignore the variability across seeds and are not adjusted for multiple comparisons.

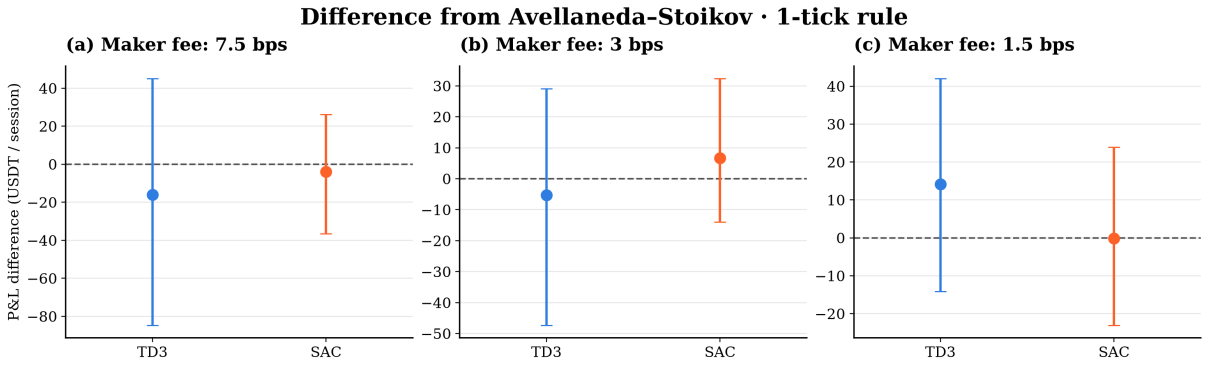


Figure 5: Paired differences between the learned policies and AS, one-tick rule. Error bars: 2.5–97.5% quantiles of a circular block bootstrap over sessions.

Figure 6 replays the same policies under the three execution rules. The rule changes the P&L more than the choice of method, and changes some rankings. With $m = 0$, TD3 earns about 50 USDT per session at 1.5 bps but loses about 220 at 7.5 bps. The effect of a stricter rule is not monotonic, since it removes adverse fills as well as profitable ones.

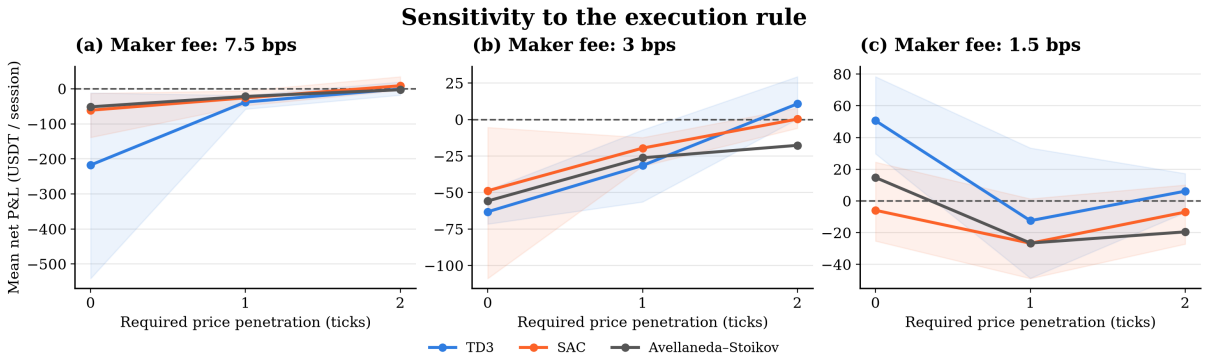


Figure 6: Mean net P&L under the zero-, one- and two-tick execution rules, with the same policies and sessions. Lines: seed means; shading: seed range. Panel scales differ.

These results are consistent with the calibration: with $1/\kappa \approx 1.6$ bps, the gain per fill is of the order of the lowest fee and about five times smaller than the highest, so profitability depends on fees and on the execution model more than on the policy. The transfer is also limited by the gap between simulated and real execution, by the change of regime between 2021–2023 and 2025–2026, and by the variance across seeds.

7 Conclusion

We embedded the Hawkes order flow of our previous work into the Cartea–Jaimungal–Ricci framework, derived the HJB equation and the optimal quotes, and rewrote the problem in event time following Guéant and Manziuk. Because the state moves between two market orders, the event-time Bellman equation involves an integral along the flow; its operator is a contraction, and its sampled version, with a random discount factor, is the target of the TD3 and SAC critics.

On 22 DOTUSDT test sessions, all methods lose money after fees under the primary execution rule. TD3 improves on Avellaneda–Stoikov by about 14 USDT per session at 1.5 bps and SAC by about 7 at 3 bps, but the bootstrap intervals contain zero and the dispersion across seeds is of the same order. The execution rule matters more than the algorithm.

The study has several limitations: one market and 22 non-consecutive test sessions, a single Hawkes calibration over periods with different regimes, no queue priority or market impact in the simulator and the replay, and only three seeds. The test sessions had also been looked at in earlier versions of the study, so the test is not fully out-of-sample and the results should be read as exploratory. One TD3 policy (seed 67, 7.5 bps) has no checkpoint satisfying the guardrails and uses the fallback, and another (seed 67, 3 bps) keeps its starting policy.

Two extensions would follow. First, a queue-reactive execution model, so that the fill probability depends on the queue position. Second, a time-varying calibration, to address the non-stationarity already observed in [28], or training directly on historical data with an offline RL objective.

A Appendix

A.1 Derivation of the HJB equation

We give the formal derivation of Proposition 3.1; see [23] for the verification arguments in the jump-diffusion setting.

Step 1 (dynamics). Between market orders the state follows the drift and diffusion of Section 3. At a market order of side $\pm$, the market state jumps from x to $T^{\pm}x$, and the inventory jumps by $\mp\Delta$ if our quote is filled (Table 1).

Step 2 (dynamic programming). For $h > 0$, split the integral on $[0, h]$ and $[h, \infty)$, and substitute $t = h + u$ in the second part:

$$\int_h^\infty e^{-rt}(\cdots)_t \,\mathrm{d}t = e^{-rh}\int_0^\infty e^{-ru}(\cdots)_{h+u}\,\mathrm{d}u.$$

Conditioning on $\mathcal{F}_h$ and using that the future depends on the past only through Y_h (Markov property) and that the dynamics do not depend on time (homogeneity), we obtain the dynamic programming principle

$$V(y) = \sup_{\mathcal{A}} \mathbb{E}_y\left[\int_0^h e^{-rt}\Big((\delta_t^+ - c)\Delta\,\mathrm{d}N_t^+ + (\delta_t^- - c)\Delta\,\mathrm{d}N_t^- + (\alpha_t q_t - \psi(q_t))\,\mathrm{d}t\Big) + e^{-rh}V(Y_h)\right].$$

Step 3 (Itô's formula). Itô's formula for jump processes gives

$$\begin{aligned} e^{-rh}V(Y_h) = V(y) &+ \int_0^h e^{-rt}(-rV + \mathcal{L}V)(Y_t)\,\mathrm{d}t \\ &+ \int_0^h e^{-rt}\sigma_\alpha\,\partial_\alpha V(Y_t)\,\mathrm{d}B_t + \sum_{0<t\le h} e^{-rt}(V(Y_t) - V(Y_{t^-})). \end{aligned}$$

Step 4 (expectations and compensators). The integrand $\sigma_\alpha\partial_\alpha V(Y_t)$ is known before the increment of B, so the Itô integral has zero expectation. The jumps occur at the jump times of $M^{\pm}$; for a predictable H, $\mathbb{E}\big[\int_0^h H_t\,\mathrm{d}M_t^{\pm}\big] = \mathbb{E}\big[\int_0^h H_t\lambda_t^{\pm}\,\mathrm{d}t\big]$ (compensator property [4]). At a buy market order the jump of V is $V(q-\Delta, T^+x) - V(q,x)$ with probability $\widetilde{h}_+(\delta^+)$ and $V(q, T^+x) - V(q,x)$ otherwise, so

$$\begin{aligned} &\mathbb{E}\Big[\sum_{t\le h,\,\Delta M_t^+ = 1} e^{-rt}(V(Y_t) - V(Y_{t^-}))\Big] \\ &\quad= \mathbb{E}\Big[\int_0^h e^{-rt}\lambda_t^+\Big(\widetilde{h}_+\left(V(q_t - \Delta, T^+x_t) - V(q_t, x_t)\right) \\ &\qquad\qquad + (1-\widetilde{h}_+)\big(V(q_t, T^+x_t) - V(q_t,x_t)\big)\Big)\,\mathrm{d}t\Big], \end{aligned}$$

and similarly for sells. In the same way,

$$\mathbb{E}\Big[\int_0^h e^{-rt}(\delta_t^+ - c)\Delta\,\mathrm{d}N_t^+\Big] = \mathbb{E}\Big[\int_0^h e^{-rt}(\delta_t^+ - c)\Delta\,\lambda_t^+\widetilde{h}_+(\delta_t^+)\,\mathrm{d}t\Big].$$

Substituting into the dynamic programming principle and subtracting $V(y)$,

$$\begin{aligned} 0 = \sup_{\mathcal{A}} \mathbb{E}_y\Big[\int_0^h e^{-rt}\Big\{&(-rV + \mathcal{L}V)(Y_t) + \alpha_t q_t - \psi(q_t) \\ &+ \sum_{\pm}\lambda_t^{\pm}\Big[\widetilde{h}_{\pm}\big((\delta_t^{\pm} - c)\Delta + \mathbb{A}^{\pm}V\big) + \mathbb{B}^{\pm}V\Big](Y_t)\Big\}\,\mathrm{d}t\Big]. \end{aligned}$$

Dividing by h and letting $h \to 0^+$, by dominated convergence (under the growth assumption) and right-continuity, the average over $[0,h]$ converges to the value of the integrand at $t=0$, where $Y_0 = y$ is deterministic. The supremum over processes becomes a supremum over the values $\delta^\pm \geq 0$ at time 0.

Step 5 (regrouping). In the bracket, $\widetilde{h}(V(q \mp \Delta, T^\pm x) - V) + (1-\widetilde{h})(V(q, T^\pm x) - V) = \widetilde{h}\,\mathbb{A}^\pm V + \mathbb{B}^\pm V$, which gives (3). The terms in q, α and ψ are deterministic given y and do not depend on δ, so the supremum only acts on $\widetilde{h}_\pm\big((\delta - c)\Delta + \mathbb{A}^\pm V\big)$, separately for each side.

A.2 Law of the drift between events

On $\{u < \tau\}$, $\mathrm{d}\alpha_u = -\zeta\alpha_u\,\mathrm{d}u + \sigma_\alpha\,\mathrm{d}B_u$. Itô's formula applied to $e^{\zeta u}\alpha_u$ gives $\mathrm{d}(e^{\zeta u}\alpha_u) = \sigma_\alpha e^{\zeta u}\,\mathrm{d}B_u$, hence

$$\tilde{\alpha}(u) = \alpha e^{-\zeta u} + \sigma_\alpha \int_0^u e^{-\zeta(u-v)}\,\mathrm{d}B_v.$$

The stochastic integral of a deterministic function is Gaussian and centred, with variance given by Itô's isometry,

$$\sigma_\alpha^2 \int_0^u e^{-2\zeta(u-v)}\,\mathrm{d}v = \frac{\sigma_\alpha^2}{2\zeta}(1 - e^{-2\zeta u}).$$

Hence

$$\tilde{\alpha}(u) \sim \mathcal{N}\Big(\alpha e^{-\zeta u}, \tfrac{\sigma_\alpha^2}{2\zeta}(1 - e^{-2\zeta u})\Big),$$

and $\mathbb{E}\int_0^\infty e^{-ru}\,|\tilde{\alpha}(u)|\,\mathrm{d}u \leq \int_0^\infty e^{-ru}(\,|\alpha| + \sigma_\alpha/\sqrt{2\zeta})\,\mathrm{d}u < \infty$ by Cauchy–Schwarz, which justifies the use of Fubini in Remark 4.6.

A.3 Proof of the contraction

Proof of Proposition 4.9. Write $\mathcal{T}W(q,x) = C(q,x) + \mathbb{E}\big[e^{-r\tau}\,\mathcal{M}W(q, \phi_\tau x) \mid x\big]$ with

$$\mathcal{M}W(q,z) = \sum_{\pm} \frac{\lambda^\pm(z)}{\bar{\lambda}(z)} \sup_{\delta \geq 0} \Big\{\widetilde{h}_\pm\big((\delta - c)\Delta + W(q \mp \Delta, T^\pm z)\big) + (1 - \widetilde{h}_\pm)W(q, T^\pm z)\Big\}.$$

For any functions f, g, $|\sup f - \sup g| \leq \sup|f - g|$. For fixed δ, the difference of the brackets for W_1 and W_2 is a convex combination, with weights $\widetilde{h}_\pm$ and $1 - \widetilde{h}_\pm$, of differences of $W_1 - W_2$, hence bounded by $\|W_1 - W_2\|_\infty$. Summing with the weights $\lambda^\pm/\bar{\lambda}$ gives $|\mathcal{M}W_1 - \mathcal{M}W_2| \leq \|W_1 - W_2\|_\infty$, and the term C cancels, so $|\mathcal{T}W_1 - \mathcal{T}W_2|\,(q,x) \leq \mathbb{E}[e^{-r\tau} \mid x]\,\|W_1 - W_2\|_\infty$.

Under Assumption 4.8, $\bar{\lambda}(\phi_v x) \leq \Lambda_{\max}$, so $\mathcal{S}_x(u) \geq e^{-\Lambda_{\max} u}$: τ stochastically dominates an $\mathrm{Exp}(\Lambda_{\max})$ variable τ_0. Since $t \mapsto e^{-rt}$ is decreasing, $\mathbb{E}[e^{-r\tau} \mid x] \leq \mathbb{E}[e^{-r\tau_0}] = \Lambda_{\max}/(r + \Lambda_{\max})$. Finally, $\mathcal{T}$ maps bounded functions to bounded functions, since $|C| \leq (|Q|\,\alpha_{\max} + \psi(Q))/r$ and the fill gains are bounded by the optimal values of Corollary 3.4. The Banach fixed-point theorem applies, and $\widetilde{V}$, which satisfies $\widetilde{V} = \mathcal{T}\widetilde{V}$ by Corollary 4.4, is the unique bounded fixed point. □

A.4 Hyperparameters

Table 5: Hyperparameters of the synthetic training and of the fine-tuning. Values in braces form the selection grid.

	Synthetic training	Fine-tuning
Hidden layers × units	2×64	from checkpoint
Parallel environments	64	64
Transitions	$\leq$ 8,388,608	1,572,864
Replay buffer	65,536	1,048,576
Batch size	256	256
Actor / critic learning rate	$3 \cdot 10^{-4}$ / 10^{-3}	$\times\{0.1, 0.25, 0.5\}$
Updates per environment step	1	$\{2, 8\}$
Polyak coefficient ϱ	0.05	0.005
TD3 exploration / target noise / clip	0.2 / 0.2 / 0.5	0.1 / 0.2 / 0.5
TD3 policy delay d	2	2
SAC initial temperature	0.02	0.001
Random initial transitions	4,096	16,384 (pretrained policy)
Critic warm-up (actor frozen)	–	131,072
Anchor weight w_0 / decay N_{anchor}	–	$\{0, 0.4\}$ / 786,432
Actor moving average	–	0.999
Evaluation every	262,144 (synthetic)	131,072 (historical)

All networks are trained with Adam [17] and gradient clipping at norm 10.

References

[1] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.

[2] Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1):1550005, 2015.

[3] Steven J. Bradtke and Michael O. Duff. Reinforcement learning methods for continuous-time Markov decision problems. In *Advances in Neural Information Processing Systems*, volume 7, pages 393–400, 1995.

[4] Pierre Brémaud. *Point Processes and Queues: Martingale Dynamics*. Springer, 1981.

[5] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995.

[6] Álvaro Cartea, Sebastian Jaimungal, and Jason Ricci. Buy low, sell high: A high frequency trading perspective. *SIAM Journal on Financial Mathematics*, 5(1):415–444, 2014.

[7] Stefan Elfwing, Eiji Uchibe, and Kenji Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, 107:3–11, 2018.

[8] Scott Fujimoto and Shixiang Shane Gu. A minimalist approach to offline reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 20132–20145, 2021.

[9] Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 1587–1596, 2018.

[10] Olivier Guéant. Optimal market making. *Applied Mathematical Finance*, 24(2):112–154, 2017.

[11] Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507, 2013.

[12] Olivier Guéant and Iuliia Manziuk. Deep reinforcement learning for market making in corporate bonds: beating the curse of dimensionality. *Applied Mathematical Finance*, 26(5):387–452, 2019.

[13] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 1861–1870, 2018.

[14] Tuomas Haarnoja, Aurick Zhou, Kristian Hartikainen, George Tucker, Sehoon Ha, Jie Tan, Vikash Kumar, Henry Zhu, Abhishek Gupta, Pieter Abbeel, and Sergey Levine. Soft actor-critic algorithms and applications. arXiv:1812.05905, 2018.

[15] Alan G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971.

[16] Thomas Ho and Hans R. Stoll. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1):47–73, 1981.

[17] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[18] Vijay R. Konda and John N. Tsitsiklis. Actor-critic algorithms. In *Advances in Neural Information Processing Systems*, volume 12, pages 1008–1014, 2000.

[19] Charles L. Lawson and Richard J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, 1974.

[20] Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2016.

[21] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

[22] Yosihiko Ogata. On Lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1):23–31, 1981.

[23] Bernt Øksendal and Agnès Sulem. *Applied Stochastic Control of Jump Diffusions*. Springer, 2nd edition, 2007.

[24] Dimitris N. Politis and Joseph P. Romano. A circular block-resampling procedure for stationary data. In Raoul LePage and Lynne Billard, editors, *Exploring the Limits of Bootstrap*, pages 263–270. Wiley, 1992.

[25] Warren B. Powell. *Approximate Dynamic Programming: Solving the Curses of Dimensionality*. Wiley, 2nd edition, 2011.

[26] Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley, 1994.

[27] David Silver, Guy Lever, Nicolas Heess, Thomas Degris, Daan Wierstra, and Martin Riedmiller. Deterministic policy gradient algorithms. In *Proceedings of the 31st International Conference on Machine Learning (ICML)*, volume 32 of *PMLR*, pages 387–395, 2014.

[28] Maxime Skryma. Modelling Bitcoin market order dynamics using a bivariate Hawkes process with a sum-of-exponentials kernel. Working paper, 2026.

[29] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

[30] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017.